

Particulate Matter

Monitoring & Health Effects

Issue 2026-10-03 | Entry window 3 October 2026 (full day) | 14 records screened in scope

14RECORDS IN SCOPE
(45 EXCLUDED)**7**OF 10 SUBTOPIC
BANDS POPULATED**5**EFFECT ESTIMATES
WITH A CI**1**TIER-A
RECORDS**4**RECORDS CARRIED ON
METADATA ALONE

Signal of the day

Constituent-specific lung-cancer risk in a quarter-million never-smokers — and the constituent it fingers is the one low-cost sensors measure worst.

Xia and colleagues (*J. Hazard. Mater.*, tier A) followed **262,785** never-smokers in the China Kadoorie Biobank to **2,646** incident lung cancers and assigned annual $\text{PM}_{2.5}$ constituents from a satellite/ML reanalysis, 2004–2018. Per IQR: nitrate **HR 1.35** (1.23–1.48), ammonium **1.22** (1.13–1.31), chloride **1.20** (1.12–1.28); the quantile g-computation mixture gives **1.13** (1.07–1.19), with NO_3^- and NH_4^+ the dominant contributors and an additive and multiplicative $\text{NO}_3^- \times \text{PRS}$ interaction. **Caveat:** nitrate and ammonium are strongly collinear in any reanalysis product and are partly inherited from the total-mass predictors, so “nitrate drives the risk” may be “the best-resolved component wins”. **Why it matters for sensing:** ammonium nitrate is semi-volatile. It evaporates in a heated inlet and in the warm optical cavity of a low-cost node, which is why nephelometric sensors under-read exactly the fraction this cohort implicates — and why a composition-blind network cannot stratify the exposure that carries the effect.

What changed today

- **Marine $\text{PM}_{2.5}$ by extrapolation.** XGBoost on AOD, meteorology and $\text{NO}_2/\text{CO}/\text{O}_3/\text{SO}_2$ gives 2019–2024 $\text{PM}_{2.5}$ fields over the eastern China seas, falling west-to-east with Bohai Bay maxima and a visible 2020 drop; SHAP puts AOD first in spring–autumn, meteorology in winter. There are almost no marine ground monitors, so training and held-out validation both lean on coastal land sites — open-water skill is unverified, and no shipping-emission predictor enters the model.
- **Waterpipe cafés as occupational environments.** Five operating venues in Al-Madinah: weekday $\text{PM}_{2.5}$ **$285.5 \pm 149 \mu\text{g m}^{-3}$** , PM_{10} **301.6 ± 156** , weekend TVOC **0.50 mg m^{-3}** , formaldehyde below reference. A $\text{PM}_{2.5}/\text{PM}_{10}$ ratio near 0.95 is the signature of pure combustion with no coarse fraction. Instrument and humidity correction unstated — but a factor-of-two optical bias still leaves these an order of magnitude above guideline.
- **A properly bounded null.** Two-sample MR of six pollutants against otitis media with effusion (12,397 cases / 464,237 controls) finds nothing surviving FDR; the nominal PM_{10} OR **1.56** (0.95–2.55) reverses with the 88-SNP set. The authors report minimum detectable ORs of **1.69–32.1**, so moderate effects are not excluded — and ambient-PM genetic instruments index residence and behaviour, not exposure.
- **Cold-start particle number.** A 1.6 L GDI engine with an electrically heated catalyst plus secondary air over the first 80 s of the cold WLTC cut THC 69.1 %, CO 65.7 % and NO_x 80.6 % for 26.7 Wh; PN rose sharply at -7°C coolant but fell $>30\%$ under coordinated control. PN is reported only as a relative change — no absolute count, size distribution or sub-23 nm fraction, which is where cold-start GDI particles sit and where optical nodes are blind.

Window continuity

The 2 October issue closed with `last_entry_date` at 2 October. This issue collects **3 October in full**, continuous with it. **Five scheduled runs (4–8 October) harvested and screened but did not author**, so dailies for 4–8 October and the W40 weekly (27 September–3 October) are still owed and follow in this backfill. One admitted record (an *ES&T* in-vitro immunotoxicology paper, 10.1021/acs.est.6c07515) could not be written in this run and is held, not rejected; it enters a later issue.

Corpus analytics

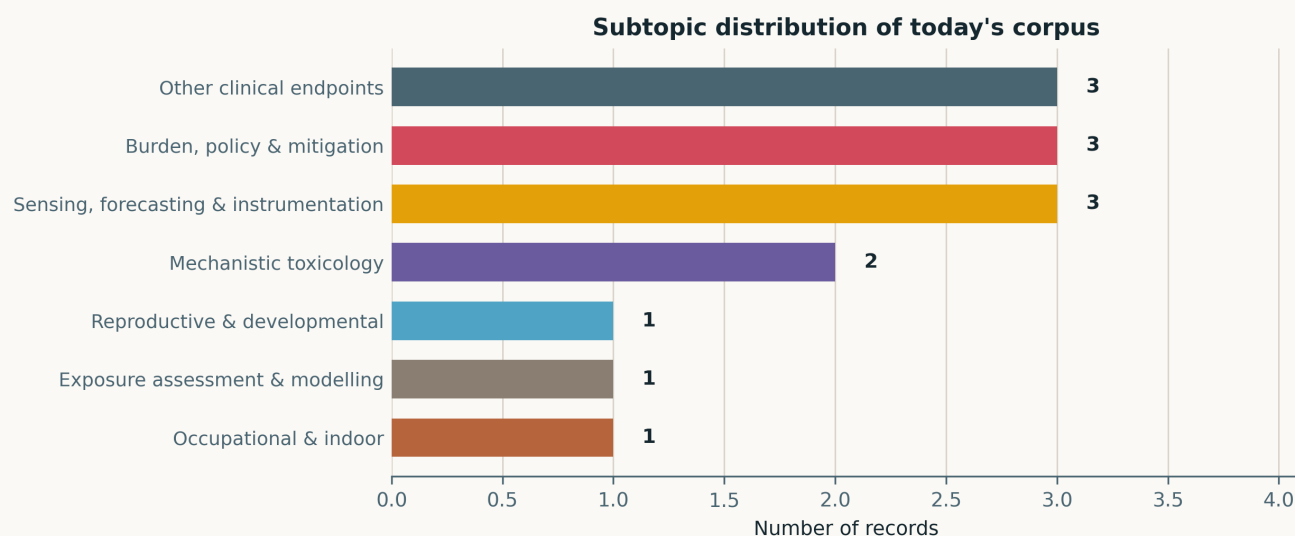


Fig. Subtopic assignment of the 14 in-scope records. **Seven of ten bands populated;** *Other clinical*, *Burden* and *Sensing* carry **three** each, *Mechanistic toxicology* two, and one each for *Reproductive*, *Exposure assessment* and *Occupational & indoor*. *Cardiovascular*, *Neuro* and *Respiratory* are empty — the day's only respiratory-adjacent record is a never-smoker lung-cancer cohort, filed under other clinical endpoints.

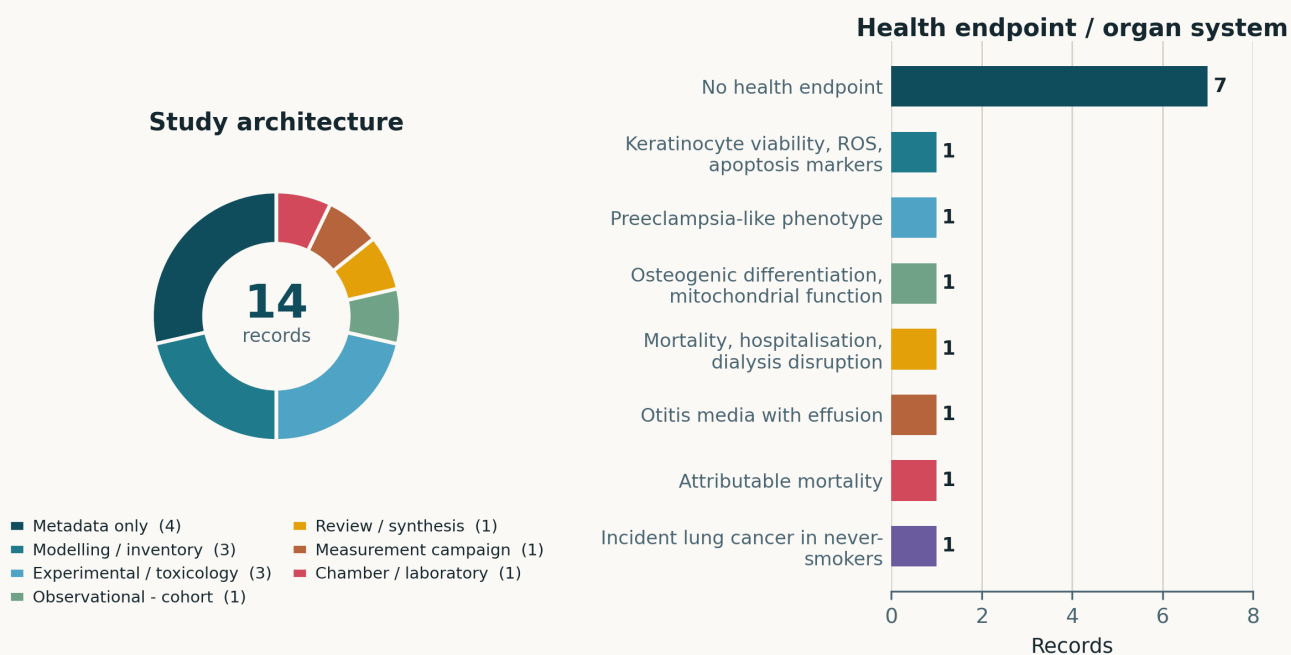


Fig. Left — architecture: **four** records carry no abstract and assert no design; *modelling / inventory* and *experimental / toxicology* hold **three** each (the machine-learning marine surface, the aviation health-impact model and the Mendelian-randomisation analysis; the mouse model and two in-vitro studies), with one each for the prospective cohort, the review, the indoor measurement campaign and the engine bench. Right — **eight** records report no health endpoint.

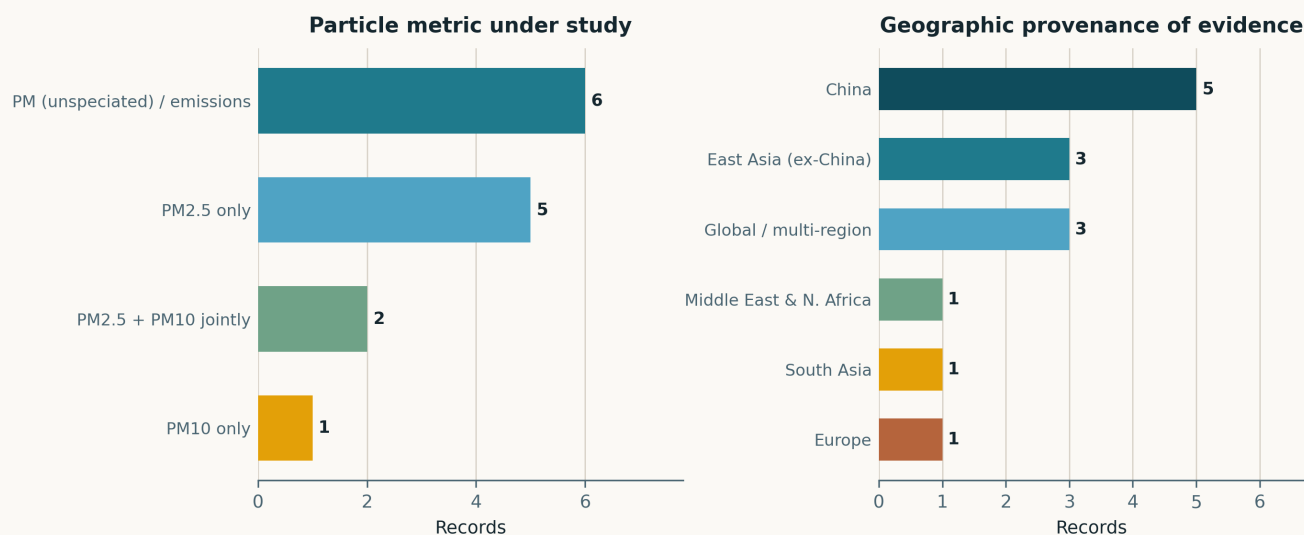


Fig. Left — particle metric: unspciated PM or emissions six (the four metadata-only deposits, the engine bench and the aviation burden), PM_{2.5} only five, PM_{2.5} + PM₁₀ jointly two (the waterpipe cafés and the MR analysis), PM₁₀ only one. The constituent cohort is filed under PM_{2.5}; no record reports particle number or composition as its primary metric. Right — geography: China five, global / not stated three, East Asia excluding China three (both Korean laboratory studies and the engine bench), and one each for Europe, South Asia and the Middle East & North Africa.

Life-course windows addressed today (records may span several stages)

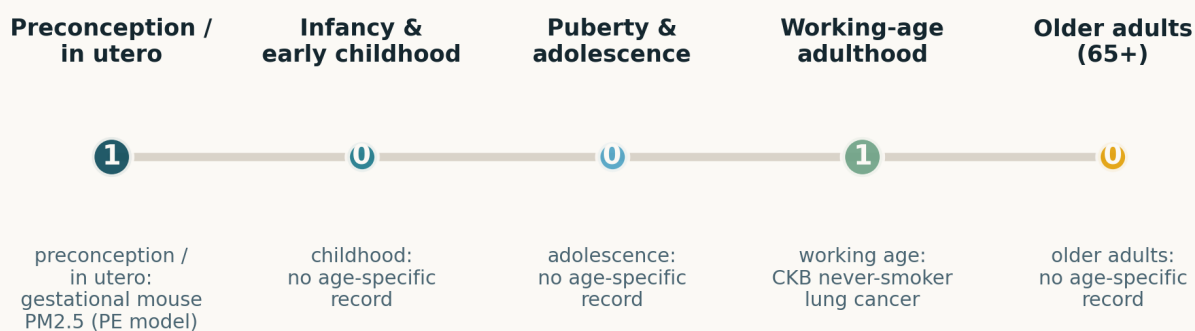


Fig. Life-course windows: in utero one (the gestational mouse preeclampsia model), working age one (the CKB never-smoker cohort, adult at baseline); childhood, adolescence and older adults have no age-specific record. The in-utero entry is experimental, not a human birth cohort.

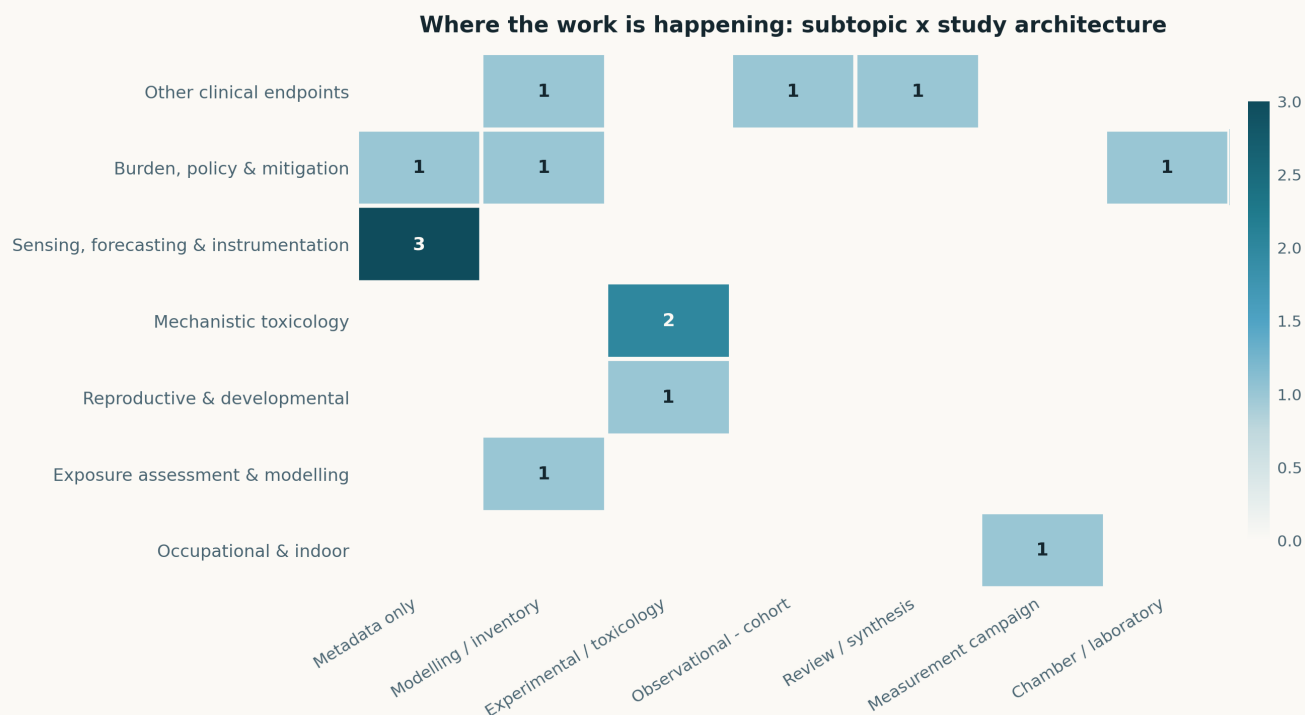


Fig. Subtopic × study architecture for the 14 records. The metadata-only block sits across sensing and burden; toxicology concentrates in mechanistic and reproductive; the measurement and modelling cells are each occupied once.

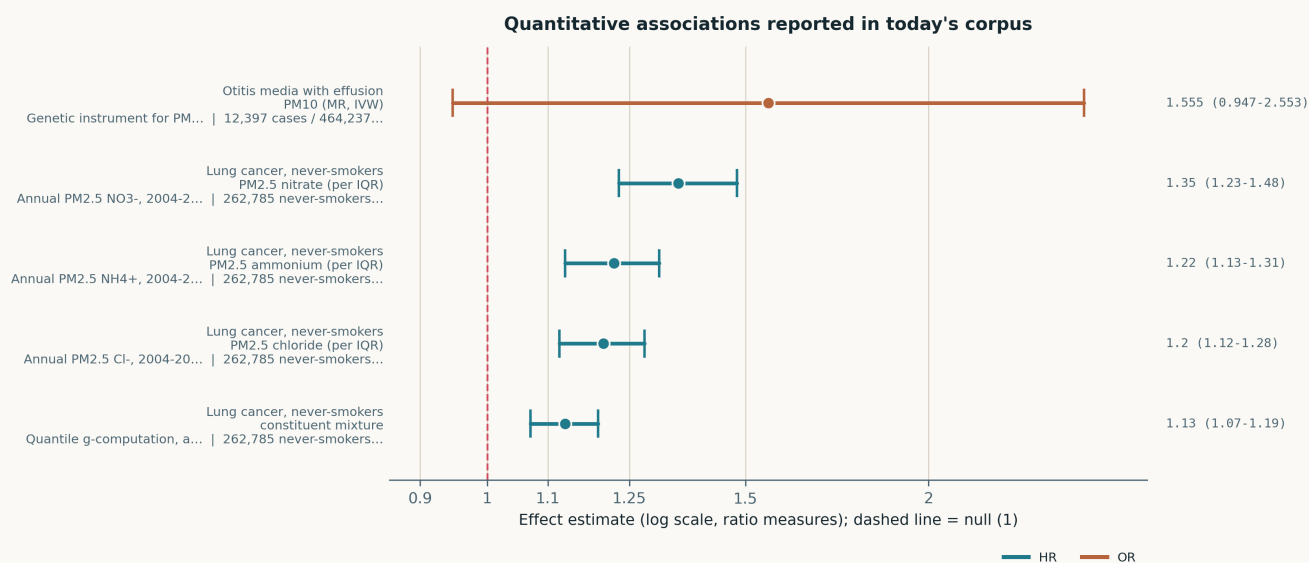


Fig. Five ratio estimates with 95% CIs, log scale. The four hazard ratios are per-IQR constituent and mixture effects from one cohort (China Kadoorie Biobank never-smokers) and share cases, so they are not poolable with each other. The fifth is the two-sample MR odds ratio for PM₁₀ and otitis media with effusion — a different estimand on a genetic instrument, plotted for scale, and it crosses unity. The aviation burden estimate is a death count, not a ratio, and does not enter the panel.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Reproductive & developmental

1 record

Ma et al. 2026 J Environ Manage

Maternal PM_{2.5} exposure induces preeclampsia-like phenotypes in mice through placental angiogenic dysfunction and lncRNA-RUNX1 dysregulation

C57BL/6J dams given intranasal heating-season PM_{2.5} through gestation developed raised blood pressure at E13.5 and E17.5, proteinuria, lower VEGF and PlGF, higher TNF- α , IL-6 and sFlt-1, and abnormal labyrinth architecture with impaired angiogenesis; ET-1 and NO were unchanged. Transcriptomics plus ChIP/RIP implicated lncRNAs NONMMUT087808.1 and NONMMUT044449.2 acting through RUNX1. Limitation: 3 mg/kg bolus intranasal instillation is far above any ambient-equivalent deposited dose and bypasses the nasal-pharyngeal filter, so the phenotype demonstrates sufficiency, not ambient relevance; one PM source, no filtered-air-plus-vehicle dose-response. Relevance: moderate - the sFlt-1/PlGF axis is measurable in human pregnancy and is the natural mediator to test against measured exposure.

doi: [10.1016/j.jenvman.2026.131066](https://doi.org/10.1016/j.jenvman.2026.131066)

Mechanistic toxicology

2 records

Zhang et al. 2026c Free Radic Biol Med

PM_{2.5} exposure impairs BMSC function and dysregulates mitochondrial oxidative stress: multi-omics insights into potential osteoporotic mechanisms

Rat BMSCs exposed to PM_{2.5} lost viability, cytoskeletal integrity, proliferation and migration, with suppressed osteogenesis via β -catenin/Runx2, raised ROS and disturbed mitochondrial homeostasis. Transcriptomics gave 2,372 DEGs (911 up / 1,461 down) and metabolomics 171 DEMs, converging on cAMP signalling; forskolin rescued viability, ROS, membrane potential, migration and differentiation. Limitation: a single pooled PM_{2.5} sample of unstated composition at unstated dose, applied directly to a cell type that in vivo sits behind the alveolar barrier - the rescue is strong internal evidence for the pathway but says nothing about delivered dose at the marrow. Relevance: low for sensing, moderate for the bone-outcome literature that epidemiology has begun to report.

doi: [10.1016/j.freeradbiomed.2026.10.004](https://doi.org/10.1016/j.freeradbiomed.2026.10.004)

Lee et al. 2026b Cell Biol Int

Curcumin-Loaded Nanospheres Attenuate Fine Dust (PM₁₀)-Induced Cellular Toxicity by Suppressing Reactive Oxygen Species-Dependent Signaling in Human Keratinocytes

PM₁₀ reduced HaCaT viability and raised ROS, ERK, p38 and NF- κ B activity, shifted Bax/Bcl-2, cleaved caspase-3 and processed IL-1 β ; curcumin-loaded nanospheres at 1e-4 μ M restored viability and blunted each step, as did NAC and pathway inhibitors. Limitation: this is a formulation paper - the PM₁₀ arm is a generic oxidative-stress stimulus with no composition, dose-response or endotoxin control, and an effect at 1e-4 μ M with none at higher unformulated doses is a non-monotonic claim that needs replication before it means anything. Relevance: low - logged for completeness of the dermal-endpoint strand, not as mechanism.

doi: [10.1002/cbin.70222](https://doi.org/10.1002/cbin.70222)

Exposure assessment & modelling

1 record

Wen and Chang 2026 Environ Pollut

Estimating marine PM_{2.5} concentrations and exploring the drivers over the eastern China seas using multi-source data and machine learning

BPNN, random forest and XGBoost compared under sample-, spatial- and temporal-held-out validation for PM_{2.5} over the Bohai, Yellow and East China seas; XGBoost best. Estimated fields fall west-to-east and north-to-south with maxima in Bohai Bay and off Shandong, a declining trend with winter maxima, a visible 2020 drop, and SHAP attributing spring-autumn variance to AOD and winter variance to meteorology. Limitation: there are almost no marine ground monitors, so training and validation both lean on coastal land sites - held-out skill over open water is unverified and the 'marine' field is an extrapolation; no shipping-emission predictor. Relevance: moderate - the obvious use case for buoy- or vessel-mounted low-cost nodes is precisely this validation gap.

doi: [10.1016/j.envpol.2026.129276](https://doi.org/10.1016/j.envpol.2026.129276)

Sensing, forecasting & instrumentation

3 records

Rawat et al. 2026 Atmos Environ*Integrated Assessment of PM_{2.5} in the Central Himalaya using Satellite, Reanalysis, and a new In-situ CUPi Sensor network*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 3 Oct with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still refusing the connection. By title this is exactly the fusion problem this watch tracks - a new in-situ sensor network (CUPi) evaluated against satellite and reanalysis PM_{2.5} in complex mountain terrain, where both gridded products are weakest. Relevance: potentially high; on the retry list, and worth a full-text read when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122395

Chen et al. 2026b J Aerosol Sci*Linking Aerosol Chemical Composition to Volatility: A Predictive Parameterization and Implications for Future Clean Air*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 3 Oct without an abstract. By title, a parameterisation predicting aerosol volatility from chemical composition. Volatility is the quantity that decides how much of an ambient aerosol survives the heated inlet of a reference monitor or the warm optical cavity of a low-cost sensor, so a composition-based predictor is directly relevant to the semi-volatile bias that separates sensor and FEM mass. Relevance: potentially high; on the retry list.

doi: 10.1016/j.jaerosci.2026.106902

Kang et al. 2026 Build Environ*Moisture-coupled aerosol filtration in moss-based porous media: enhanced particle capture and aerodynamic resistance trade-off*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 3 Oct with no abstract. By title, a filtration study of moss-based porous media in which moisture improves particle capture at a cost in aerodynamic resistance. The efficiency-vs-pressure-drop trade-off is the same constraint that governs inlet and filter design in gravimetric samplers and in filtered-air intervention trials. Relevance: low-moderate; on the retry list.

doi: 10.1016/j.buildenv.2026.115363

Occupational & indoor

1 record

Aljabri et al. 2026 Indoor Air*Indoor Air Quality and Exposure Characterization in Operational Waterpipe Cafes: Field Measurements of Particulate and Volatile Pollutants in Al-Madinah Al-Munawwarah*

Direct measurement in five operating waterpipe cafes on weekdays and weekends: mean weekday PM_{2.5} 285.5 +/- 149 µg m⁻³ and PM₁₀ 301.6 +/- 156 µg m⁻³, weekend TVOC 0.50 mg/m³, formaldehyde below short-term reference values. Variability tracked smoking intensity and ventilation. The PM_{2.5}/PM₁₀ ratio near 0.95 is the signature of a pure combustion source with no coarse fraction. Limitation: instrument make, calibration and whether an optical sensor was humidity- or composition-corrected are not stated - charcoal and tobacco smoke is exactly where optical PM monitors read high; five venues, no personal sampling, no worker shift-length exposure. Relevance: high for occupational indoor monitoring, where a factor-of-two optical bias still leaves concentrations an order of magnitude above guideline.

doi: 10.1155/ina/8494753

Burden, policy & mitigation

3 records

Quadros et al. 2026 Commun Earth Environ*Global air quality and human health impacts of growing aircraft emissions*

Global aircraft emissions projected 2019-2040 and run through a chemistry transport model. 2019: 33,900 (95% CI 23,500-45,600) PM_{2.5} deaths and 24,600 (15,500-34,200) O₃ deaths attributable to aviation. By 2040 these rise 58%/82% (low), 119%/148% (baseline) and 169%/205% (high scenario); the non-aviation background shifts aircraft impacts by up to 44%. Limitation: the CIs propagate concentration-response uncertainty only, not CTM, inventory (especially cruise NOx-nitrate) or grid resolution, which dominate a marginal-sector estimate; most PM_{2.5} impact is secondary nitrate and sulfate far from airports. Relevance: moderate - the near-airport UFP signal low-cost networks can see is not the mass-based burden counted here.

doi: 10.1038/s43247-026-03732-4

Lee et al. 2026c ACS Omega*Exhaust Emission Characteristics of Gasoline Engines with Secondary Air Systems and Electrically Heated Catalysts in Cold-Start WLTC Mode*

A 1.6 L GDI engine with warm-up and underfloor catalysts, an electrically heated catalyst (EHC) and secondary air injection (SAI) was run over the cold WLTC phase, including a -7 C coolant case. EHC + SAI over the first 80 s cut THC 69.1%, CO 65.7% and NOx 80.6% for 26.7 Wh; best strategy was EHC pre-heat 50 s before start, held 30 s after. SAI without EHC raised CO. PN rose sharply in the cold case but optimised EHC-SAI coordination cut it by >30%. Limitation: one engine, one calibration, dynamometer cycle rather than real driving, and PN is reported only as a relative reduction - no absolute PN, size distribution or sub-23 nm fraction, which is where cold-start GDI particles sit. Relevance: moderate - cold-start PN is the largest remaining tailpipe contribution in urban low-cost-network data, and sub-23 nm particles are invisible to the optical nodes watching for it.

doi: 10.1021/acsomega.6c05980

Zhang et al. 2026d Atmos Pollut Res*Impact of strong biomass burning in Northeast China: air pollution, enhancement ratio among pollutants, and health risk*

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 3 Oct with no abstract. By title, an episode study of strong agricultural biomass burning in Northeast China reporting enhancement ratios among co-emitted pollutants and a health-risk estimate. Enhancement ratios (dPM / dCO, for instance) are the quantity that lets a sparse network attribute an episode to burning rather than to secondary formation, so the numbers matter more than the burden claim. Relevance: moderate; on the retry list.

doi: 10.1016/j.apr.2026.103224

Other clinical endpoints

3 records

Xia et al. 2026 J Hazard Mater*Long-term exposure to fine particulate matter constituents and lung cancer risk in Chinese never-smokers*

262,785 never-smokers in the China Kadoorie Biobank, 2,646 incident lung cancers; time-dependent Cox models per IQR of each constituent, quantile g-computation for the mixture, and a polygenic risk score for interaction. HR per IQR: nitrate 1.35 (1.23-1.48), ammonium 1.22 (1.13-1.31), chloride 1.20 (1.12-1.28); mixture HR 1.13 (1.07-1.19), dominated by NO₃⁻ and NH₄⁺; additive and multiplicative NO₃⁻ × PRS interaction. Limitation: constituents come from one satellite/ML reanalysis in which nitrate and ammonium are strongly collinear and partly inherited from total-mass predictors, so 'nitrate drives the risk' may be 'the best-resolved component wins'; second-hand smoke and cooking fuel only partly captured. Relevance: high - constituent-specific risk is the argument for composition-capable nodes (nitrate is semi-volatile and biased low on heated optical sensors).

doi: 10.1016/j.jhazmat.2026.143759

Yang and Liu 2026 medRxiv (preprint)*No evidence for large causal effects of six air pollutants on non-suppurative otitis media: a two-sample Mendelian randomization study*

Two-sample MR with UK Biobank pollutant GWAS and FinnGen R13 OME (12,397 cases, 464,237 controls); IVW primary, full pleiotropy battery, positive (asthma, rhinitis, CRS) and negative (suppurative OM) controls. No pollutant survived FDR; the nominal PM₁₀ OR 1.56 (0.95-2.55) reversed with the 88-SNP set (0.64, 0.24-1.69). Minimum detectable ORs 1.69-32.1, so moderate effects are not excluded - the authors say so. Limitation: 'genetic instruments for ambient PM' index residential and behavioural traits, not exposure; null MR here is uninformative by construction, and MVMR F < 10. Relevance: low - included as a properly bounded null, not as evidence of absence.

doi: 10.64898/2026.09.30.26364391

Sandal et al. 2026 Clin Kidney J*Disasters and hazard exposure in end-stage kidney disease: a scoping review of patient outcomes*

PRISMA scoping review: 10,040 screened, 30 studies, ~1.07 million dialysis patients and 64,531 transplant recipients. Poor air quality was associated with 5-139% higher mortality; extreme heat 9-45%; 23-59% of patients had dialysis disruption in disasters. Limitation: scoping design, no pooling or risk-of-bias grading, and 'poor air quality' spans wildfire smoke, PM_{2.5} increments and AQI categories, so the 5-139% range is not an effect size. Relevance: low-moderate - dialysis patients are an identifiable, routinely contacted population where sensor-triggered alerts could plausibly be tested.

doi: 10.1093/ckj/sfag318

Corpus register

First author	Focus	Journal	Design	Metric	Region
Reproductive & developmental					
Ma et al.	Maternal PM _{2.5} exposure induces preeclampsia-like phenotypes in mice through placental angiogenic dysfunction and lncRNA-RUNX1 dysregulation	<i>J Environ Manage</i>	Animal model	Heating-season northern-China PM _{2.5} , intranasal 3 mg/kg every other day, GD0.5-17.5	China (laboratory; Taiyuan-type heating-season PM)
Mechanistic toxicology					
Lee et al.	Curcumin-Loaded Nanospheres Attenuate Fine Dust (PM ₁₀)-Induced Cellular Toxicity by Suppressing Reactive Oxygen Species-Dependent Signaling in Human Keratinocytes	<i>Cell Biol Int</i>	In vitro exposure	PM ₁₀ applied to HaCaT keratinocytes; curcumin nanospheres as the test article	South Korea (laboratory)
Zhang et al.	PM _{2.5} exposure impairs BMSC function and dysregulates mitochondrial oxidative stress: multi-omics insights into potential osteoporotic mechanisms	<i>Free Radic Biol Med</i>	In vitro exposure	PM _{2.5} suspension applied to rat bone-marrow mesenchymal stem cells (dose not in abstract)	China (laboratory)
Occupational & indoor					
Aljabri et al.	Indoor Air Quality and Exposure Characterization in Operational Waterpipe Cafes: Field Measurements of Particulate and Volatile Pollutants in Al-Madinah Al-Munawwarah	<i>Indoor Air</i>	Field measurement	Indoor PM _{2.5} and PM ₁₀ with TVOC and formaldehyde, peak occupancy, 5 venues	Saudi Arabia (Al-Madinah)
Sensing, forecasting & instrumentation					
Chen et al.	Linking Aerosol Chemical Composition to Volatility: A Predictive Parameterization and Implications for Future Clean Air	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Metadata only - composition-to-volatility parameterisation (by title)	Not recoverable from metadata
Kang et al.	Moisture-coupled aerosol filtration in moss-based porous media: enhanced particle capture and aerodynamic resistance trade-off	<i>Build Environ</i>	Metadata only (no abstract)	Metadata only - filtration efficiency vs pressure drop in moss media under humidity (by title)	South Korea (by author affiliation; not asserted)
Rawat et al.	Integrated Assessment of PM _{2.5} in the Central Himalaya using Satellite, Reanalysis, and a new In-situ CUPi Sensor network	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - satellite + reanalysis + a new in-situ low-cost PM sensor network (by title)	India (Central Himalaya)
Exposure assessment & modelling					
Wen and Chang	Estimating marine PM _{2.5} concentrations and exploring the drivers over the eastern China seas using multi-source data and machine learning	<i>Environ Pollut</i>	Machine-learning model	Marine PM _{2.5} from AOD + meteorology + NO ₂ /CO/O ₃ /SO ₂ , 2019-2024	China (eastern China seas)
Burden, policy & mitigation					
Lee et al.	Exhaust Emission Characteristics of Gasoline Engines with Secondary Air Systems and Electrically Heated Catalysts in Cold-Start WLTC Mode	<i>ACS Omega</i>	Engine dynamometer testing	Particle number (PN) with CO, THC, NO _x over the cold phase of the WLTC	South Korea (laboratory)
Quadros et al.	Global air quality and human health impacts of growing aircraft emissions	<i>Commun Earth Environ</i>	Health impact model	Aviation-attributable PM _{2.5} and O ₃ from a global CTM, 2019 and 2040 scenarios	Global
Zhang et al.	Impact of strong biomass burning in Northeast China: air pollution, enhancement ratio among pollutants, and health risk	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - biomass-burning episode, pollutant enhancement ratios, health risk (by title)	China (Northeast)

First author	Focus	Journal	Design	Metric	Region
Other clinical endpoints					
Sandal et al.	Disasters and hazard exposure in end-stage kidney disease: a scoping review of patient outcomes	<i>Clin Kidney J</i>	Scoping review	Poor air quality among heat, disaster and conflict hazards (metric varies by study)	Global (30 studies, 10 countries/regions)
Xia et al.	Long-term exposure to fine particulate matter constituents and lung cancer risk in Chinese never-smokers	<i>J Hazard Mater</i>	Prospective cohort	Annual PM _{2.5} and constituents (NO ₃ ⁻ , NH ₄ ⁺ , Cl ⁻ , SO ₄ ²⁻ , BC, OM), ChinaHighAirPollutants 2004-2018	China (China Kadoorie Biobank, 10 regions)
Yang and Liu	No evidence for large causal effects of six air pollutants on non-suppurative otitis media: a two-sample Mendelian randomization study	<i>medRxiv (preprint)</i>	Two-sample Mendelian randomisation	Genetic instruments for PM _{2.5} , PM _{2.5} absorbance, PM _{2.5} -10, PM ₁₀ , NO ₂ , NO _x (UK Biobank GWAS)	Europe (UK Biobank exposure; FinnGen R13 outcome)

Provenance and method

Entry window **3 October 2026**, single day, continuous with the 2 October issue. Harvest and screening for this day were completed on the **6 October** run and cached (`cache/2026-10-03/_fresh.json`); this run authored from that cache without re-querying, so the per-source counts are as logged then: **PubMed**, **Europe PMC**, **Crossref by ISSN** and the **PubMed connector** ([EDAT] 3–5 October, 17 PMIDs merged across three days) contributing **60** fresh candidates after cross-day deduplication; **OpenAlex** HTTP 429; **arXiv** no entry dated 3 October. **14 admitted**, **45 rejected** with logged reasons, **1 held** to a later issue. **Four records carried on metadata alone** (29%, a series high) — all four are Elsevier or Wiley deposits with no abstract, and the abstract-retry leg recovered none of them. **Consensus** returned 10 calibration hits, **0 new**. **Scholar Gateway** fails with `ACCESS_DENIED` and has now been unavailable for eleven days. **ClinicalTrials.gov**: EPIC-AIR (NCT07500948, Leicester, $n=80$, real-time PM_{2.5}/PM₁₀/NO₂ guidance in cardiac and pulmonary rehabilitation) is new in the window and enters the trial watch. DOI gate: **0 fail**, **0 warn**. Abstracts remain the copyright of their respective publishers.